- For states with discharge-level data:
    - We calculated age-, health endpoint-, and county-specific hospitalization counts. South Carolina was the only state that, while not providing discharge-level data, did provide county-level data for each age group-endpoint combination.
    - The above calculation excluded hospitalizations with missing patient age or county FIPS, which may lead to underestimation of rates. Therefore, we scaled up the previously calculated age-, endpoint-, and county-specific counts using an adjustment factor obtained as follows:
        - We first counted the number of discharges for a specific endpoint in the state including those discharges with missing age or county FIPS.
        - We then counted the number of discharges for the endpoint in the state excluding those records with missing age or county FIPS.
        - The adjustment factor is the ratio of the two counts.
    - For California and West Virginia, patient county was unavailable for all observations. For these two states, we used hospital county in place of patient county.
    - For health outcomes deemed acute (acute myocardial infarction; cerebrovascular events; stroke; pneumonia; lower respiratory infection; acute cases of asthma), we distributed patients within the hospital state in cases where the patient resided out of state. We assume that everyone admitted to the hospital in a given state developed that acute condition while in that state.
    - We calculated hospitalization rates for each county by dividing the adjusted county-level hospitalization counts by the Census estimated county-level population for the corresponding year (2011 - 2014). Following CDC Wonder, we treated rates as "unreliable" when the hospitalization count was less than 20, using the same procedure we used for mortality rates (see Section D.1.1).

  For states with summarized state statistics (from HCUPnet) we calculated the state-, age-, endpoint- specific hospitalization rates and applied them to each county in the state. We used the previously described procedure to adjust the "unreliable" rates.
- For states without discharge-level or state-level data:
    - We obtained the endpoint-specific hospitalization counts in each region from HCUPnet/NIS (we refer to this count for the ith endpoint in the jth region as "TOTALij")

- For those states in the jth region that do have discharge-level or state-level data, we summed the hospital admissions by endpoint (we refer to this count for the ith endpoint in the jth region as “SUB ij”).
- We then estimated the hospitalization count for states without discharge or state data for the ith endpoint in the jth region as TOTALij - SUB ij. Note that while this count is endpoint- and region- specific, it is not age-specific. We obtained the distribution of hospital admission counts across age groups based on the Central Distributor data and assumed the same distribution for the HCUPnet hospitalizations. We then applied this distribution to the estimated hospital counts (i.e., TOTALij - SUB ij) to obtain endpoint-, region-, and age-specific counts.
- Using the corresponding age- and region-specific populations in BenMAP-CE from Woods and Poole (2015), we calculated age-specific hospitalization rates for the ith endpoint in the jth region and applied them to those counties in the region that didn’t have discharge-level or state-level data.

The endpoints in hospitalization studies are defined using different combinations of ICD codes. Rather than generating a unique baseline incidence rate for each ICD code combination, for the purposes of this analysis, we identified a core group of hospitalization rates from the studies and applied the appropriate combinations of these rates in the health impact functions:

- congestive heart failure (ICD-9 428)
- dysrhythmia (ICD-9 427)
- heart rhythm disturbances (ICD-9 426-427)
- acute myocardial infarction (ICD-9 410)
- ischemic heart disease - 1 (ICD-9 410-414)
- ischemic heart disease - 2 (ICD-9 410-414, 429)
- ischemic heart disease (less myocardial infarction) (ICD-9 411-414)
- all cardiovascular (ICD-9 390-429)
- all cardiovascular (less myocardial infarctions) (ICD-9 390-409, 411-429)
- cardiovascular, cerebrovascular and peripheral vascular diseases (ICD-9 410-414, 429, 426-427, 428, 430-438, 440-449)
- all cardiac outcomes (ICD-9 390-459)

- cerebrovascular events (ICD-9 430-438)
- stroke (ICD-9 431-437)
- peripheral vascular disease - 1 (ICD-9 440-448)
- peripheral vascular disease - 2 (ICD-9 440-449)
- all respiratory (ICD-9 460-519)
- respiratory illness - 1 (ICD-9 466, 480-486, 490-493)
- respiratory illness -2 (ICD-9 464-466, 480-487, 490-492)
- chronic lung disease (ICD-9 490-496)
- chronic lung disease (less asthma) (ICD-9 490-492, 494-496)
- chronic lung disease (less asthma) -2 (ICD-9 490-492, 494, 496)
- chronic lung disease (less asthma) -3 (ICD-9 490-492)
- chronic lung disease (less asthma) -4 (ICD-9 491,492, 494, 496)
- pneumonia (ICD-9 480-486)
- asthma (ICD-9 493)
- lower respiratory infection (ICD-9 466.1, 466.0, 480-487, 490, 510-511)
- respiratory – 1 (ICD-9 491, 492, 493, 496)
- respiratory – 2 (ICD-9 464-466, 480-487, 490-492, 493)
- alzheimer’s disease (ICD-9 331.0)
- parkinson’s disease (ICD-9 332)

In addition to the hospitalization endpoints above, we developed a set of county level baseline incidence for one EHA endpoint, All Respiratory (see Section E.7.8 for epidemiological description). We generated the EHA rates by applying the HCUPnet national ratio of All Respiratory hospitalizations originating from the emergency department (77%) to the county level incidence rates developed from the discharge and state-level data.

For each C-R function, we selected the baseline rate or combination of rates that most closely matches to the study endpoint definition. For studies that define chronic lung disease as ICD 490- 492, 494-496, we subtracted the incidence rate for asthma (ICD

493) from the chronic lung disease rate (ICD 490-496). In some cases, the baseline rate will not match exactly to the endpoint definition in the study. For example, Burnett et al. (2001) studied the following respiratory conditions in infants <2 years of age: ICD 464.4, 466, 480-486, 493. For this C-R function we apply an aggregate of the following rates: ICD 464, 466, 480-487, 493. Although they do not match exactly, we assume that relationship observed between the pollutant and study-defined endpoint is applicable for the additional codes. Table D-10 presents a summary of the national hospitalization rates for 2014 from HCUP.

Table D-10. Hospitalization Rates (per 100 people per year), by Health Endpoint and Age

Hospitalization Category	ICD-9 Code	Age 0-1	2-17	18-24	25-34	35-44	45-54	55-64	65-74	75-84	85+
Respiratory											
All Respiratory	460-519	2.387	0.363	0.166	0.212	0.340	0.737	1.297	2.292	4.151	6.343
Pneumonia	480-486	0.477	0.101	0.039	0.063	0.103	0.196	0.336	0.640	1.426	2.660
Chronic Lung Disease	490-496	0.226	0.151	0.041	0.056	0.105	0.281	0.496	0.837	1.276	1.306
Asthma	493	0.217	0.147	0.036	0.048	0.076	0.123	0.136	0.157	0.218	0.243
Cardiovascular											
All Cardiovascular	390-429	0.044	0.017	0.061	0.138	0.377	0.914	1.747	3.131	5.886	8.832
Acute Myocardial Infarction, Nonfatal	410	0.000	0.000	0.002	0.010	0.068	0.202	0.380	0.575	0.921	1.332
Ischemic Heart Disease	410-414	0.000	0.000	0.002	0.014	0.105	0.350	0.689	1.090	1.570	1.734
Dysrhythmia	427	0.016	0.005	0.014	0.025	0.057	0.145	0.319	0.684	1.357	1.917
Congestive Heart Failure	428	0.010	0.001	0.005	0.021	0.061	0.165	0.344	0.700	1.727	3.513
Stroke	431-437	0.009	0.003	0.007	0.021	0.070	0.199	0.417	0.816	1.639	2.488
Neurological											
Alzheimer’s Disease	331.0	0.000	0.000	0.00	0.00	0.00	0.0004	0.0035	0.027	0.129	0.248
Parkinson’s Disease	332	0.000	0.000	0.00011			0.0037		0.020		0.025

D.3 Nonfatal Heart Attacks

The relationship between short-term particulate matter exposure and heart attacks was quantified in a case-crossover analysis by Peters et al. (2001). The study population was selected from heart attack survivors in a medical clinic. Therefore, the applicable population to apply to the C-R function is all individuals surviving a heart attack in a given year. Several data sources are available to estimate the number of heart attacks per year. For example, several cohort studies have reported estimates of heart attack incidence rates in the specific populations under study. However, these rates depend on the specific characteristics of the populations under study and may not be the best data to extrapolate nationally. The American Heart Association reports approximately